

Week 5 Worksheet 1 Solutions

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Exercise 1. On the homework, you will show that the matrix elements of x and p in the harmonic oscillator eigenbasis $\{|n\rangle\}$ are given by

$$x_{mn} = \langle m|x|n\rangle = \sqrt{\frac{\hbar}{2m\omega}}(\delta_{m,n+1}\sqrt{n+1} + \delta_{m,n-1}\sqrt{n})$$

$$p_{mn} = i\sqrt{\frac{m\omega\hbar}{2}}(\delta_{m,n+1}\sqrt{n+1} - \delta_{m,n-1}\sqrt{n}).$$

- a) For finite-dimensional matrices, show that $\text{tr}([A, B]) = 0$, where for an $(n \times n)$ -matrix A , the **trace** $\text{tr}(A)$ is the sum of the diagonal entries of A .

Hint: Write $\text{tr}(AB)$ in summation notation as $\sum_{i,j} c_{ij}$; that is, determine the scalars c_{ij} in terms of the matrix elements of A and B .

- b) *Paradox.* From the result of (a), it would seem that the matrices which you will compute on the homework imply that $\hbar = 0$, since $[x, p] = i\hbar$. Use the matrix elements x_{mn} and p_{mn} to calculate xp and px , and explain in detail why the paradoxical conclusion $\hbar = 0$ is invalid.

- a) Given a matrix A with entries a_{ij} , $\text{tr}(A) = \sum_i a_{ii}$. Since the matrix entries of AB are $a_{ik}b_{kj}$, where a_{ij} and b_{ij} are the entries of A and B , respectively, $\text{tr}(AB) = \sum_{i,j} a_{ij}b_{ji}$. On the other hand, $\text{tr}(BA) = \sum_{i,j} b_{ij}a_{ji} = \sum_{i,j} a_{ji}b_{ij} = \sum_{i,j} a_{ij}b_{ji} = \text{tr}(AB)$, where in the second-to-last equality we swap the dummy indices $i \leftrightarrow j$.

- b) We first calculate xp and px .

$$(xp)_{ij} = \frac{i\hbar}{2} \left(\delta_{i,j+2}\sqrt{i(j+1)} - \delta_{i,j}\sqrt{ij} + \delta_{i,j}\sqrt{(i+1)(j+1)} - \delta_{i,j+2}\sqrt{(i+1)j} \right),$$

where we use the sum over the middle index to collapse Kronecker deltas of the form $\delta_{i,n+1}\delta_{n,j+1} = \delta_{i,j+2}$, for example. Similarly,

$$(px)_{ij} = \frac{i\hbar}{2} \left(\delta_{i,j+2}\sqrt{i(j+1)} - \delta_{i,j}\sqrt{(i+1)(j+1)} + \delta_{i,j}\sqrt{ij} - \delta_{i,j+2}\sqrt{(i+1)j} \right).$$

Subtracting $(xp)_{ij} - (px)_{ij}$, we find

$$[x, p]_{ij} = \frac{i\hbar}{2} \left(-2\delta_{i,j}\sqrt{ij} + 2\delta_{i,j}\sqrt{(i+1)(j+1)} \right).$$

Summing this over $i = j$, we get what looks like an infinite sum of the form $\sum_n (-1)^n$, which doesn't converge! This is because the operators x, p are "unbounded" which means we can't take their trace. A symptom of this is the fact that the eigenfunctions of x and p , which are δ -functions and plane waves, respectively, are not normalizable. For example, if we look at $[x, p] = i\hbar$, we see that on the RHS we have the *infinite-dimensional* identity operator, which is also unbounded! Indeed, if we took its trace, we get ∞ , since $\sum_{n=0}^{\infty} 1 = \infty$. So there is no paradox: The trace of both sides is undefined and can't be computed; that's why we get what appears to be a non-converging sum for the trace of $[x, p]$.